DDR PHY ARCHITECTURE & CALIBRATION

HIGH-DENSITY ENGINEERING REFERENCE • SIGNALS, DQS LEVELING & VREF TRAINING

1. ESSENTIAL DDR SIGNAL DEFINITIONS

SIGNAL	FULL NAME	DIRECTION	FUNCTIONAL DESCRIPTION
`DQ`	Data Input/Output	Bidirectional	Data bus carrying active payload bits (0s/1s) to/from DRAM chips.
`DQS / DQS#`	Data Strobe (Diff Pair)	Bidirectional	Source-synchronous clock specifying exact edge alignments for latching DQ.
`CS#`	Chip Select	Input (DRAM)	Enables specific memory rank/device to decode command/ address bus.
`DM / DBI#`	Data Mask / Bus Inversion	Input (DRAM)	Masks write byte lanes or manages signal power/noise via data inversion.

2. DQS WRITE LEVELING (TIME-DOMAIN CALIBRATION)

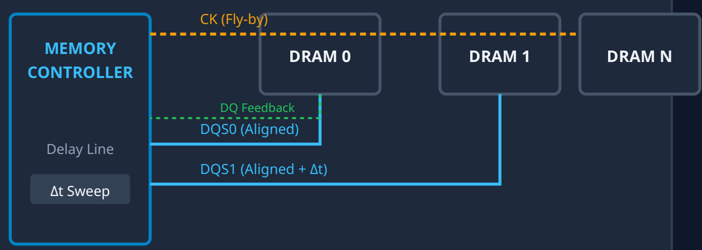
Problem: Fly-By Skew

Modern modules use **Fly-By topology** for Clock (CK) and Command/Address (CA) to preserve high-speed signal integrity. This introduces propagation delays across ranks, creating temporal skew between `CK` and `DQS` at individual DRAM devices.

Calibration Protocol

- **Init:** Controller enables Write Leveling mode in DRAM Mode Register (MR1).
- **Sweep:** Controller increments DQS delay line in fine steps.
- **Feedback:** DRAM samples `CK` using `DQS` edge and returns value on `DQ`.
- **Lock:** Delay locked at 0-to-1 transition, aligning `DQS` edge with `CK`.

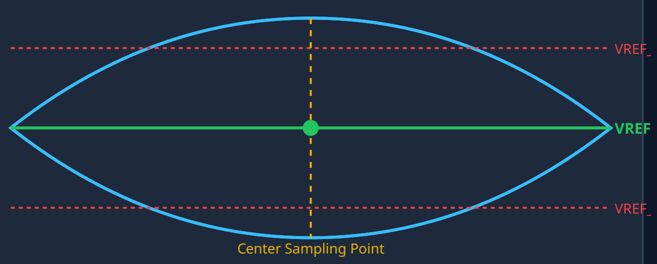
FLY-BY TOPOLOGY & WRITE LEVELING FLOW



3. VREF TRAINING (VOLTAGE-DOMAIN OPTIMIZATION)

VREF calibration optimizes the vertical DC voltage threshold to establish maximum noise margin at the center of the Data Eye. Driven by asymmetric signaling shifts in modern protocols.

DATA EYE & 2D CENTERING (TIME VS. VOLTAGE)



2D Eye Search Algorithm

1. **Sweep:** Increment internal `VrefDQ` via Mode Register commands.
2. **Measure Window:** Test valid timing window width (t_{DIV}) at each voltage step.
3. **Boundaries:** Identify V_{REF_MAX} and V_{REF_MIN} pass/fail thresholds.
4. **Lock Midpoint:** Program $V_{REF\_OPT} = rac{V_{REF\_MAX} + V_{REF\_MIN}}{2}$.

4. GENERATIONAL TECHNOLOGY COMPARISON

FEATURE / PARAMETER	DDR3	DDR4	DDR5
I/O Signaling Standard	SSTL (Push-Pull)	POD12 (Pseudo Open Drain)	POD11 / Low Power POD
VREF Source	External Board Divider	Internal DRAM Generator (VrefDQ)	Per-Receiver Internal LDOs
Target VREF DC Level	Symmetrical (~50% V_{DDQ})	Asymmetrical (~70–80% V_{DDQ})	Variable / Dynamic Optimization
Calibration Granularity	None (Fixed Board Baseline)	Per-Channel / Per-DRAM Rank	Per-Pin / Per-Lane Fine Calibration
CA VREF Support	No	Yes (VrefCA Training)	Yes (Per-channel independent VrefCA)